

EXAM 7 – SPRING 2013

17. (2.75 points)

Given the following data:

	<u>Policy Year Ultimate Loss (\$000)</u>				
<u>Line of Business</u>	<u>2008</u>	<u>2009</u>	<u>2010</u>	<u>2011</u>	<u>2012</u>
Personal Auto	900	1,150	6,000	850	1,100
Homeowners	340	200	1,060	150	250

a. (1 point)

Calculate the Pearson product-moment correlation between the personal auto loss data and the homeowners loss data.

b. (1 point)

Calculate the Spearman rank correlation between the personal auto loss data and the homeowners loss data.

c. (0.75 point)

Explain the difference between these two measures of correlation as applied to this data set.

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Question 17 Sample Answer

Solution 1

$$\begin{aligned}\text{a) Pearson } \rho &= \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}} \\ &= \frac{3298500}{\sqrt{20065000 \times 564200}} \\ &= 0.98035\end{aligned}$$

	PA		HOME	$X_i - \bar{X}$	$Y_i - \bar{Y}$
	900		340	-1100	-60
	1150		200	-850	-200
	6000		1060	4000	660
	850		150	-1150	-250
	1100		250	-900	-150
$\bar{X}$	2000	$\bar{Y}$	400		

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 3298500$$

$$\sum (X_i - \bar{X})^2 = 20065000$$

$$\sum (Y_i - \bar{Y})^2 = 564200$$

Solution 2

$$\begin{aligned}\text{a) } \rho &= \frac{E(XY) - E(X)E(Y)}{\sigma_X \sigma_Y} \\ &= \frac{1459700 - 2000 \times 400}{(2003.25)(335.92)} \\ &= 0.98\end{aligned}$$

$$E(X) = 2000$$

$$E(Y) = 400$$

$$E(XY) = 1,459,700$$

$$\sigma_X = 2003.25$$

$$\sigma_Y = 335.92$$

Solution 1

$$b) \rho = 1 - \frac{6 \times \sum d_j^2}{N(N^2 - 1)} = 1 - \frac{8}{5(5^2 - 1)/6} = 0.60$$

X	Y	Rank X	Rank Y	Diff ²
900	340	2	4	4
1150	200	4	2	4
6000	1060	5	5	0
850	150	1	1	0
1100	250	3	3	0
				8

Solution 1

- c) Pearson's correlation is calculated based on values, while Spearman's correlation is based on ranks. Thus, Pearson's correlation can be impacted by large outlier amounts. This impact can be seen here, where the 2010 year is largely influencing the correlation factor in the Pearson calculation. The use of ranks in Spearman's calculation is not impacted by this, so while it shows some positive correlation with 0.6, it is not nearly as high as the Pearson result.

Solution 2

- c) Pearson is a cardinal measure & is driven by the magnitude of the values. It will be highly leveraged by points far away from the mean. This occurs in 2010 which both Auto & HO have losses above the mean. This point drives the high correlation factor of 0.98. Spearman is an ordinal measure, weighted equally among all 5 years.

Examiner Comment

Part a), this is an application of the formulas for Pearson's correlation. The syllabus material provides two approaches to calculate the correlation using the provided data. Common errors include arithmetic errors and mixing two formulas together.

Part b), this is an application of the formulas for Spearman's rank correlation. The syllabus material provides a formula and approach for estimating the correlation using the provided data. Common errors include arithmetic errors and memorizing the incorrect formula.

Part c), common errors include only give generic statements of difference without tying back to the data given in the question.